



MCU-Friendly Mamba
Optimizing Mamba Model Inference on Microcontrollers
with Quantization and Low-Rank Factorization

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An electronic version of this thesis is available at <http://repository.tudelft.nl/>.

Abstract

The abstract should concisely summarize the motivation, research question, methodology, key findings, and conclusions of the paper. It should clearly explain why optimizing mamba inference is important, what experimental framework was developed, and what the main findings were.

The abstract should be understandable as a standalone text and should avoid unnecessary references or detailed technical descriptions.

1 Introduction

The section should:

- Introduce embedded AI and TinyML.
- Explain deployment constraints such as latency, memory, and computational limitations.
- Briefly explain Mamba model potential.
- Review the most relevant existing literature.
- Identify the research gap.
- Clearly state the research question.
- Summarize the main contributions of the work.
- Briefly explain the structure of the paper.

The introduction should remain accessible to readers with a general computer science background.

2 Related Work and Background

This section should provide the theoretical and research background necessary to understand the paper.

Possible subsections include:

2.1 Embedded AI and TinyML

Discuss:

- TinyML and edge AI
- Quantization techniques
- Embedded deployment constraints

2.2 Mamba Architecture

Discuss:

- What is a Mamba model
- How it utilizes SSM
- What are its main advantages and challenges

2.3 Research Gap

Explain:

- Why existing work is insufficient
- Why memory optimizations are necessary
- Why deployment-aware evaluation is important

This section should support the motivation for the experimental framework introduced later in the paper.

3 Methodology

This section should describe the experimental framework and explain how the research question is investigated.

3.1 Dataset and Preprocessing

Describe:

- Dataset selection

3.2 Experimental Design

Describe:

- Quantization strategies
- Memory optimization strategies
- Ways of measuring used memory
- Testing strategies and reliability
- Repeated runs and statistical stability

3.3 Model Architecture

Describe:

- Changes made to the original
- Hyperparameter choices

3.4 Deployment Optimization

Describe:

- Deployment choices
- INT8, INT4 quantization
- Quantization-aware training
- Latency benchmarking

4 Embedded Deployment Setup

This section should describe the embedded-oriented deployment environment and evaluation setup.

4.1 Software Stack

Describe:

- No platform IO
- No ONNX
- C library
- Pytorch
- Brevitas
- Torchao
- Other relevant libraries

4.2 Deployment Measurements

Explain:

- Model size measurement
- Memory throughout runtime measurement
- Latency benchmarking methodology
- Inference execution setup

4.3 Reproducibility

Describe:

- Random seeds
- Repeated runs
- Configuration consistency
- Code availability

5 Results

This section presents the experimental results of the impact of different techniques on the accuracy and memory usage. Results should be supported by descriptive and clear figures.

5.1 Effect of Quantization on Memory and Accuracy

This subsection presents:

- Memory reductions across quantization strategies
- Accuracy changes induced by quantization
- Variance of performance across repeated runs

5.2 Embedded Efficiency Trade-offs

This subsection presents:

- Inference latency measurements
- Model size comparisons
- Accuracy-efficiency relationships

5.3 Effects of other memory optimizations on accuracy and latency

This subsection presents:

- Impacts of Low rank factorization

6 Responsible Research

This section discusses the ethical implications of the work, reproducibility considerations, and the role of AI-assisted tools in the development process.

6.1 Ethical Considerations

Key ethical aspects include:

- Possibility of reducing power usage
- Enabling bad actors to run dangerous ML models on embedded devices (listening)

6.2 Reproducibility

The following aspects support reproducibility:

- Publicly available dataset and deterministic preprocessing pipeline
- Ease of reproducing the model training environment
- Access to the microcontroller version used
- Consistent evaluation metrics across all experimental runs
- Code structured to enable rerunning of full experimental pipelines

6.3 Use of AI Tools

Artificial intelligence tools were used as supportive instruments during the development of this work. Their use was limited to improving code readability, assisting with debugging, and refining written text for clarity and grammatical correctness.

Specifically, AI assistance was used for:

- Debugging implementation issues in experimental code
- Improving clarity and structure of written explanations
- Assisting with formatting and consistency in LaTeX writing

All scientific decisions, experimental design choices, and interpretation of results were made independently by the author. AI tools were not used to generate or alter experimental outcomes, results, or conclusions.

7 Discussion

This section interprets the experimental findings and discusses their implications for embedded trustworthy AI systems. It also discusses the limitations of the approach.

7.1 Trade offs between accuracy, memory efficiency, and latency

Discuss:

- How making models smaller reduces accuracy of them
- How some optimizations can be made that hurt latency

7.2 How Mamba model performs

Discuss:

- The performance of Mamba compared to SOTA for the dataset

7.3 Limitations

Discuss:

- Dataset limitations
- More quantization possible
- Hardware specific implementation
- Difficulty of comparison with other models
- Lack of standardized tooling

8 Conclusions and Future Work

This section should directly answer the research question and summarize the main findings.

The conclusion should:

- Restate the research objective
- Summarize the most important findings
- Explain the memory limitations of Mamba models on MCUs
- Highlight key trade-offs identified in the experiments

Future work may discuss:

- Other memory optimization strategies

- More standardized deployment (LiteRT custom operation)
- Energy-aware optimization

The conclusion should remain concise and readable as a standalone section.

A Some further guidelines that go without saying (right?)

- Follow the required page limit and formatting constraints (maximum 8 pages, two-column format).
- Ensure that the paper clearly communicates the central research goal and contributions.

A.1 Reference use

- Use a numbered citation style (IEEE format), where references appear as [1], [2]...
- Use BibTeX or a similar tool to automatically generate bibliographic information.
- Every citation in the text must appear in the bibliography, and no uncited references should be included.
- All ideas, figures, and data derived from other work must be properly referenced.
- Literal quotations must include quotation marks and page numbers where applicable.
- Paraphrased content must be sufficiently distinct from the original source.
- Plagiarism is strictly prohibited.

A.2 Structure

Paragraphs

- are well-constructed and logically organized
- are not too long: each paragraph discusses a single idea
- start with clear topic sentences
- contribute to a coherent flow from research question to conclusions
- support claims with appropriate references and evidence

A.3 Style

- Use a formal academic tone throughout the paper
- Prefer “we” over “I” when referring to the authors
- Avoid informal language and subjective phrasing
- Avoid unsupported or overly strong claims
- Be precise and cautious in wording (e.g., do not claim “privacy achieved” when only proxies are used)
- Ensure correct use of English: clear grammar, no spelling or lexical errors
- Maintain clarity: avoid overly complex or ambiguous sentences
- Use varied sentence structure with a mix of active and passive voice

A.4 Tables and figures

- All tables and figures must be numbered (e.g., Figure 1, Table 1)
- Each must include a clear and descriptive caption
- Every table and figure must be referenced at least once in the text
- Figures and tables must be interpretable independently of the main text
- Avoid overly dense or cluttered visualizations
- If adapted or copied, include proper references

A.5 Page limit and formatting

- The paper must not exceed 8 pages in two-column format
- Use consistent font, spacing, and layout throughout the document
- Follow a standard conference-style formatting
- Prioritize clarity and conciseness over excessive detail

A.6 Key focus of the paper

- The paper must clearly demonstrate how embedded deployment constraints (such as model size and latency) reshape memory–accuracy–latency trade-offs in AI systems
- All sections should support and contribute to this central theme

References

- [1] F. Lastname, *example reference*, 1900.